

Question:

There exist real numbers  $x$  and  $y$ , both greater than 1, such that  $\log_x(y^x) = \log_y(x^{4y}) = 10$ . Find  $xy$ .

Reasoning + Answer:

We are given that there exist real numbers  $x$  and  $y$ , both greater than 1, such that :

$$\log_x(y^x) = \log_y(x^{4y}) = 10$$

We are to find  $xy$ .

Step 1 : Simplify the first equation

$$\log_x(y^x) = 10$$

Using logarithmic identities :

$$\log_x(y^x) = x \log_x y = 10$$

So :

$$x \log_x y = 10 \quad \text{(1)}$$

Step 2 : Simplify the second equation

$$\log_y(x^{4y}) = 10$$

$$\log_y(x^{4y}) = 4y \log_y x = 10$$

So :

$$4y \log_y x = 10 \quad \text{(2)}$$

Step 3 : Use change of base

We know that :

$$\log_x y = \frac{1}{\log_y x}$$

Let's define  $a = \log_y x$ , so  $\log_x y = \frac{1}{a}$

Now plug into equations (1) and (2):

From equation (1):

$$x \cdot \frac{1}{a} = 10 \Rightarrow x = 10a \quad \text{(3)}$$

From equation (2):

$$4y \cdot a = 10 \Rightarrow y = \frac{10}{4a} = \frac{5}{2a} \quad \text{(4)}$$

Step 4 : Use the relation between  $x$  and  $y$  through logarithms

We know  $a = \log_y x$ , so by definition :

$$a = \frac{\log x}{\log y}$$

But we also have expressions for  $x$  and  $y$  in terms of  $a$  :

$$\text{From (3): } x = 10a$$

$$\text{From (4): } y = \frac{5}{2a}$$

So plug into the logarithmic expression :

$$a = \frac{\log(10a)}{\log\left(\frac{5}{2a}\right)}$$

Let's write this equation :

$$a = \frac{\log(10a)}{\log\left(\frac{5}{2a}\right)}$$

We now solve this equation for  $a$ .

Step 5 : Expand logarithms

Use logarithmic identities :

Numerator :

$$\log(10a) = \log 10 + \log a = 1 + \log a$$

Denominator :

$$\log\left(\frac{5}{2a}\right) = \log 5 - \log 2 - \log a = \log\left(\frac{5}{2}\right) - \log a$$

Let's compute  $\log\left(\frac{5}{2}\right)$ , but we can keep it symbolic.

So :

$$a = \frac{1 + \log a}{\log\left(\frac{5}{2}\right) - \log a}$$

Let's define  $b = \log a$ , so  $\log a = b$ ,  $a = e^b$  (but maybe not needed yet).

So :

$$a = \frac{1 + b}{\log(5/2) - b}$$

Now, since  $a = e^b$ ? Not necessary. Just substitute  $a$  in terms of  $b$  :

But we want to eliminate  $a$ . So from above :

$$a(\log\left(\frac{5}{2}\right) - b)$$